The Story of YatrAI

A Narrative Journey Through Chennai's Commute: Concept Proposal

Document Purpose:	Final Solution Architecture & Walkthrough
Case Study Context:	Daily commute from SRMIST (Kattankulathur) to Chennai Central & OMR
Target Audience:	Developer Team, UX Designers, Hackathon Partners
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CONFIDENTIAL CONCEPT PROPOSAL

This document details the production-feasible architecture of YatrAI, resolving core engineering frictions including geofenced ticketing, hybrid Time-Dependent A* (TD-A*) routing, and MTC bus schedule modifiers.

Chapter 1: The Daily Transit Puzzle (The Conflict)

Every morning, **Priya**, a student living on the **SRMIST Kattankulathur** campus, faces a frustrating travel challenge. Today, she has a double mission: first, she must go to **Chennai Central** to drop off a project package, and then head to her internship office at **OMR**.

To accomplish this, Priya does not use a single app. Instead, her phone screen is a chaotic dashboard of overlapping services. She opens **Google Maps** to figure out the fastest transfers. She opens the **UTS App** to purchase a local train ticket—but it errors out, refusing to issue a ticket due to GPS location jitter placing her inside the CRIS 30-meter track-proximity safety perimeter while walking near the dorms near the station. She switches to **Where Is My Train** to check if the local train is delayed. She opens **Chalo** to search if there is a connecting local MTC bus from Guindy. Finally, she opens **Rapido or Uber** to estimate how much she will have to pay for the final auto ride, only to find a 40% surge price.

Priya is mentally exhausted before she even steps past the SRM gate. She has to buy three separate tickets, coordinate transfer timelines manually, and risk getting lost at massive transit stations.

Chapter 2: The Conversational Voice (The Solution)

Instead of shuffling between five apps, Priya opens a single interface: **YatrAI**. She presses the microphone button and speaks naturally in Tanglish (Romanized Tamil):

> "SRM main gate la irundhu Chennai Central ku ponum. Appram anga irundhu OMR ponum. Budget la kaatunga."
> ("I need to go from SRM main gate to Chennai Central, and then to OMR. Show me a budget-friendly option.")

Inside YatrAI, the system resolves the query. We avoid latency bottlenecks (15-30s LLM loading spinners) by using a **hybrid execution flow**. The system parses her intent using a single LLM, instantly calculates the mathematical transit legs using a **Time-Dependent A* (TD-A*)** algorithm on local coordinates in under 10ms, and coordinates the agents in parallel to structure the final briefing:

> **Suburban Train Agent:** "I have a suburban train leaving Potheri Station at 8:15 AM. It gets to Guindy at 8:50 AM. The fare is Rs. 10."
> **Metro Agent:** "Excellent. Guindy Metro CMRL has a train departing every 5 minutes. The ride to Chennai Central takes 18 minutes. The fare is Rs. 40."
> **Last-Mile Agent:** "I checked the transfer point. The walk from the Guindy train platform to the Metro turnstiles is 150 meters. I will direct her to walk via the covered pedestrian skywalk to avoid the street traffic."
> **Master Planner Agent:** "Synthesized. Total duration is 58 minutes. Total cost is Rs. 50. This is 1/6th the cost of a private cab and completely bypasses road traffic."

Chapter 3: Choose Your Vibe (The 4 Travel Moods)

YatrAI presents Priya with four clear, dynamic journey cards tailored to her mood and constraints:

Travel Mood	Priya's Priority	The AI Route Proposal	Est. Arrival
[Cheapest]	Spend as little as possible.	Local train from Potheri to Guindy, MTC Bus 21G to Central, then MTC Bus 102 to OMR. Cost: Rs. 35.	9:45 AM
[Fastest]	Arrive quickly, ignore cost.	Rapido bike to Airport Metro, Metro to Central, then Rapido bike directly to OMR. Cost: Rs. 260.	8:50 AM
[Greenest]	Save carbon emissions.	Suburban train to Guindy, walk the skywalk, Metro to Central, then Metro to Tidel Park (OMR). Saves 2.2 kg CO_2. Earns 90 Green Points.	9:05 AM
[Safest]	Maximum safety, any time.	Pre-paid auto to Potheri, train transit, walk via CCTV skywalk, Metro transit, and a vetted cab drop to OMR. Safety Rating: 9.8/10.	9:06 AM

Chapter 4: The Ground Reality (How the Unified Pass Adapts)

Priya doesn't need to print out paper tickets. Depending on which route she chooses, the **YatrAI Unified Pass** on her screen adapts dynamically to fit the ground reality of Chennai's existing transit infrastructure:

Scenario A: Priya Chooses the [Cheapest] Route (Train + MTC Bus + Walk + MTC OMR Bus)

• **Leg 1: Suburban Rail (Potheri to Guindy)**
Priya boards the local train. The app displays an active **Suburban Pass** showing a security watermark and an animated train graphic. She shows this to the TTE on the train. [Fare: Rs. 10 | Time: 35 mins]
• **Leg 2: Transfer to Bus**
Priya exits Guindy Suburban station and crosses to the MTC bus stop. [Time: 4 mins]
• **Leg 3: MTC City Bus (Guindy to Central)**
She boards MTC Bus 21G. Since conductors will reject visual screens, Priya uses YatrAI's camera tool to scan the conductor's static UPI QR card pasted on the bus. YatrAI releases the Rs. 10 fare from her YatraWallet via a UPI proxy. The conductor's device prints her physical ticket (or on older buses, Priya performs a direct payment to the conductor's personal QR code card and receives a standard ticket). [Fare: Rs. 10 | Time: 32 mins]
• **Leg 4: Package Drop at Central**
Priya walks 200m into Central Station to drop her package at the Chennai Central station lockers. [Time: 5 mins]
• **Leg 5: MTC OMR Bus (Central to OMR)**
She boards MTC Bus Route 102 at Central bus bay. She scans the conductor's UPI QR card on board to pay Rs. 15. [Fare: Rs. 15 | Time: 40 mins]
Summary: Cost: Rs. 35 | Time: 120 mins | CO_2 Saved: 2.2 kg

Scenario B: Priya Chooses the [Fastest] Route (Rapido Bike + Metro + Rapido OMR)

• **Leg 1: Doorstep Pickup (SRM Gate to Airport Metro)**
Priya meets the Rapido rider. She shares the ride PIN **'9201'**. The rider drops her at Airport Metro. [Fare: Rs. 110 | Time: 20 mins]
• **Leg 2: Metro (Airport to Chennai Central)**
Priya enters Airport Metro. She scans the CMRL-compatible QR code at the turnstile gate and rides the Metro to Central. [Fare: Rs. 40 | Time: 22 mins]
• **Leg 3: Package Drop & Exit**
She exits Central Metro, drops her package at the Chennai Central station lockers. [Time: 5 mins]
• **Leg 4: Rapido to OMR Office**
She walks to the station exit where a second Rapido bike is waiting (auto-booked by YatrAI 3 minutes before her Metro arrived). If no rider accepts within 60 seconds, YatrAI automatically switches to an Ola Auto booking or displays the Metro continuation to Tidel Park as an alternative. She shares the ride PIN **'4891'** and rides straight to OMR. [Fare: Rs. 110 | Time: 18 mins]
Summary: Cost: Rs. 260 | Time: 65 mins | CO_2 Saved: 0.8 kg

Scenario C: Priya Chooses the [Greenest] Route (Train + Skywalk + Metro + OMR Metro)

• **Leg 1: Suburban Rail (Potheri to Guindy)**
Priya travels on the local train using the animated UTS-style pass. [Fare: Rs. 10 | Time: 35 mins]
• **Leg 2: The Skywalk Transfer**
YatrAI alerts: 'Take Platform 1 stairs and follow the covered skywalk directly into the Metro.' [Time: 3 mins]
• **Leg 3: Metro Leg (Guindy to Chennai Central)**
She scans her CMRL QR Code at the turnstile and rides to Central to drop her package at the Chennai Central station lockers. [Fare: Rs. 40 | Time: 18 mins]
• **Leg 4: Metro to OMR (Central to Tidel Park)**
Priya boards the CMRL Blue Line towards Tidel Park. Her pass remains active. She scans out at Tidel Park. [Fare: Rs. 30 | Time: 20 mins]
• **Why Metro is Greener:** MTC buses run on diesel and produce **15g CO_2/km** per passenger in heavy traffic. CMRL Metro runs on 100% electric power sourced from solar grids (at only **8g CO_2/km**). Although Metro costs more, it is significantly greener.
Summary: Cost: Rs. 90 | Time: 80 mins | CO_2 Saved: 2.2 kg (Earns 90 Green Points)

Scenario D: Priya Chooses the [Safest] Route (Vetted Auto + Train + Metro + Vetted Cab)

• **Leg 1: Secure Doorstep Pickup**
A vetted auto-driver is waiting at the campus gate. She shares the OTP to begin the trip. [Fare: Rs. 40 | Time: 5 mins]
• **Leg 2: Suburban Rail**
Priya travels on the train using the animated UTS-style pass. [Fare: Rs. 10 | Time: 35 mins]
• **Leg 3: Well-Lit Hub Transfer**
YatrAI alerts: 'Take the pedestrian skywalk at Guindy. It is CCTV-monitored, well-lit, and bypasses unlit road lanes.' [Time: 3 mins]
• **Leg 4: Metro Transit**
She scans her CMRL QR code at the turnstiles and rides the Metro to Central to drop the package at the Chennai Central station lockers. [Fare: Rs. 40 | Time: 18 mins]
• **Leg 5: Vetted Cab (Central to OMR Destination)**
At Central exit Gate 2, a pre-paid vetted cab is waiting. She shares the OTP to begin the final leg to OMR. [Fare: Rs. 120 | Time: 20 mins]
Summary: Cost: Rs. 210 | Time: 81 mins | Safety Index: 9.8/10

Chapter 5: Proactive Intelligence & Disruption Handling

A travel companion must anticipate delays and act *before* the commuter gets stuck. YatrAI handles schedule breakdowns dynamically:

Scenario 1: The Proactive Morning Call (Before Leaving)

Priya has an exam at 9:30 AM in Central. YatrAI runs in the background, syncs with her calendar, and actively monitors the transit network. At **7:45 AM**, before she has even picked up her bag, her phone sends a push notification:
"Leave in 12 minutes to catch the 8:15 AM Potheri Train and reach Central by 9:15 AM. Weather is clear, connections are on time."

[ALERT] Mid-Journey Disruption: The 8:15 AM Train Delay

At 8:02 AM, while Priya is walking towards the SRM campus exit gate, the **Disruption Agent** detects that the 8:15 AM suburban train from Potheri is delayed by 25 minutes due to a signal failure at Chengalpattu. Her 8:55 AM Metro connection is broken.

What Priya Sees & Hears:
Her phone sends an urgent double-vibrate alert. A voice prompt in Tamil/English speaks: "Alert: The 8:15 AM train is delayed. I have recalculated a faster route. Walk to the MTC Bus Stand instead."

The Automated Rerouting (Bus Schedule + Speed Matrix):
MTC buses do not consistently publish GPS coordinates. To solve the cold-start problem, YatrAI ingests MTC's open static GTFS database (sourced from the Chennai Unified Metropolitan Transport Authority - CUMTA) and overlays it with a **live road-traffic speed index** along GST Road. If the traffic speed drops to 12 km/h, the algorithm automatically adds a 15-minute delay modifier to the static bus schedule.
YatrAI calculates that MTC Bus 500C is passing the gate soon. It routes her:
1. **MTC Bus 500C** (arriving in 3 mins) Tambaram. [Fare: Rs. 15 | Time: 15 mins]
2. **Tambaram Fast Local Train** (leaves Platform 3 at 8:32 AM) Guindy. [Fare: Rs. 10 | Time: 20 mins]
3. **Metro Leg** Guindy Metro to Central. [Fare: Rs. 40 | Time: 18 mins]
Outcome: Priya reaches Central at 9:25 AM, safely on time.

[ALERT] Emergency Mid-Journey Reroute: Halting Between Stations

At 8:42 AM, while Priya is onboard the Suburban Train between Kattankulathur and Tambaram, the train unexpectedly halts due to a grid failure on the tracks. The train is delayed indefinitely.

What Priya Sees & Hears (The Haptic & Audio Warning):
YatrAI uses low-power client-side sensors. When it detects that Priya's speed drops to 0 km/h for more than 4 minutes outside a station corridor, it triggers a strong vibration pulse. Her headphones play a Tamil/English alert: "Warning: Indefinite rail delay detected. Next station (Tambaram) is 200m ahead. Recommend exiting the train at the platform when safe. Walk to the Tambaram Bus Stand."

The Pivot Options:
YatrAI displays a simplified 2D map showing the shortest walking route to the Tambaram bus depot. Simultaneously, the **Disruption Agent** checks the MTC GTFS timetable and GST Road traffic speed. Rather than abandoning her mission, YatrAI asks: "Your train is halted. I can route you directly to OMR, or via Tambaram to Central first to complete your package drop, arriving 22 minutes later. Which do you prefer?"
Priya selects the Central option. The app prepares the pre-paid UPI transaction payload for the bus conductor's static QR code card for the detour MTC bus to Central, automatically adjusting her YatraWallet escrow balance.

Chapter 6: Unified Payments, Vetting Logic & Fallbacks

This section details the core operational backend that makes YatrAI credible and friction-free on the ground.

1. The YatraWallet Escrow & Budgeting System

Priya loads money into her **YatraWallet** via UPI (Google Pay / PhonePe) in 1 tap. She sets a monthly commute budget of Rs. 2,000. When her balance falls below Rs. 150, the app sends a reminder. When she books a journey, YatrAI calculates the total fare and holds it in **escrow**.As she travels, the wallet automatically releases the split fare to each operator:
• Rs. 10 is released to the UTS ticket purchase flow when she completes the pre-filled payment screen via the UTS deep-link.
• Rs. 40 is released to CMRL Metro when she scans her screen at the Metro turnstile.
Priya never fumbles at turnstiles or payment flows.

2. Safety Vetting Logic for [Safest] Autos

To ensure the credibility of the *Safest* route, YatrAI does not book random street autos. It operates via a direct API integration with the **SRM Campus Auto-Union Registry** and ride-aggregators. Autos flagged as 'Safest' must meet strict vetting rules: the driver must have a verified police background check, the auto must have an active, GPS-linked safety tracking device, and maintain a rating above **4.8 stars** with zero safety disputes. Drivers are also selected from local women-commuter advocacy programs (such as Chennai's *Pink Patrol* network).

3. UTS Rail Ticket Fallback (TTE Disputes)

While YatrAI generates an animated digital pass matching UTS specifications, some suburban Ticket Examiners (TTEs) may refuse to accept third-party screen layouts. To prevent disputes and fines, YatrAI includes a **fail-safe fallback**:
Upon booking, YatrAI automatically registers the transaction onto the official **Indian Railways UTS digital ledger**. Priya instantly receives an official SMS ticket confirmation directly from Indian Railways sender ID ('**IR-TICKET**') on her phone. If a TTE disputes the app screen, she simply displays this government SMS, which is legally recognized as a valid ticket.

4. Green Points Redemption Model

Green Points are not decorative. Priya can redeem her points in the **Green Rewards Hub**:
• **100 Points** = Rs. 10 discount voucher applied directly to her YatraWallet balance for Metro rides.
• **200 Points** = Rs. 25 discount voucher for Rapido last-mile connections.
This creates a tangible financial incentive for commuters to choose sustainable transit.

5. Station Transfer Navigation

Large stations like Guindy and Central are confusing. To prevent user abandonment, YatrAI includes **Simplified Spatial Maps**. The app displays a clean 2D layout of the station highlighting stairs, skywalks, and turnstiles. As she walks, the app uses haptic pulses (phone vibrations) and directional arrows (e.g. 'Turn right towards Exit Gate 3') to guide her through the transfer points.

6. Keep Guard SOS Trigger Logic

When Priya selects the **Safest Route**, the app automatically generates a temporary web tracking link and sends it to her designated emergency contacts via SMS. Priya has full control: she can manually revoke access with a single tap, and the tracking link automatically expires 15 minutes after her final ride reaches the destination.

Chapter 7: Resolving Critical Engineering Frictions

To ensure that YatrAI stands up to rigorous technical questioning from hackathon judges, the architecture resolves critical real-world friction points:

Identified Friction	The Practical Reality / Constraint	YatrAI Architectural Solution
	CRIS does not provide public UTS APIs to third parties. UTS also blocks purchases if the user is sitting too close to the tracks (violating the 30-meter proximity safety rule) or too far away.	**Intent-Driven Deep Link & Geofencing:** YatrAI generates an automated Intent Payload. When Priya taps 'Book All', the app purchases the Metro ticket via open gateways, and triggers a deep-link that launches the official UTS app on her phone, pre-filling Potheri to Guindy. Local low-power OS geofencing perimeters register a circular region around the station and wake the application to trigger a lock-screen notification when she crosses the boundary to complete the payment.
MTC Bus Ticketing & UPI Proxy	MTC bus conductors will reject visual app screens. Additionally, private developers have no access to CUMTA transit ticketing databases.	**Conductor UPI Cashless Proxy:** Rather than using an unverified pass, Priya uses YatrAI's camera tool to scan the static UPI QR card pasted inside the MTC bus. YatrAI releases the fare from YatraWallet to execute the UPI transaction. On routes with ETM handheld devices, the device prints a physical ticket; on older bus routes, it defaults to a direct payment to the conductor's personal QR with a screenshot receipt.
	Running five sequential LLM calls to plan a route creates a 15–30 second loading spinner, causing users to uninstall the app.	**Hybrid Time-Dependent A* (TD-A*) Execution:** We do not use LLMs for pathfinding. Instead, route calculations are computed instantly in **under 10ms** using a custom TD-A* graph solver on local transit schedules. The LLM is used **strictly in parallel** for the initial voice-parsing of intent and the final Tamil/English text explanation, keeping total app response latency **under 1.5 seconds**.

	Unlike the Metro, MTC city buses do not consistently publish public, high-fidelity real-time GPS coordinates.	**GTFS-Static + Speed Matrix:** YatrAI ingests MTC's open static GTFS timetables (sourced from CUMTA / OpenStreetMap historical databases) and overlays it with a **live road-traffic speed index** along GST Road. If the traffic speed drops, the algorithm automatically adds a delay modifier to the static bus schedule. This is augmented by passive crowdsourcing from other active YatrAI users.

Chapter 8: Competitive Landscape & Uniqueness Matrix

Judges and peer reviewers will naturally compare YatrAI to established transit and mapping services in India. Here is a breakdown of how YatrAI acts as a **Unified Orchestrator**, eliminating the app-switching fatigue of existing siloed tools:

App / System	Mode Focus	Core Limitation for Priya	How YatrAI Integrates & Solves It
Google Maps	General routing	Static timelines; does not explain manual transfer steps or display safety indexes.	Coordinates transfers step-by-step (e.g., Guindy Skywalk guide).
Where Is My Train	Suburban/Rail	Tracks train tracks only; completely isolates buses, metro, and auto connections.	Treats rail as one leg in a unified, multi-modal itinerary.
UTS App	Rail ticketing	Frequent GPS errors near stations; requires manual ticket purchase per trip.	Generates automated Intent Payload for pre-filled UTS booking; presents unified pass.
Chalo App	Local Buses	Siloed to MTC; doesn't know Metro timetables or last-mile costs.	Combines MTC bus stops with local train and Metro transfer points.
Rapido / Uber / Ola	Private rides	Expensive for full commutes; subject to heavy traffic delays.	Uses them strictly for short, cost-effective first/last-mile connections.
RedBus / IRCTC	Intercity travel	Optimized for long-distance; unusable for daily city transfers.	Specifically optimized for intra-city daily transfers (Chennai-wide).

Conclusion & Vision:
By integrating time-dependent scheduling models, client-side geofencing, and contactless payment gateways, YatrAI transitions public transit from a series of fragmented connections into a single, cohesive travel experience. This project bridges the gap between digital accessibility and physical transport infrastructure, laying a production-ready roadmap for the future of urban smart mobility in India. Priya reached OMR at 9:20 AM, spent Rs. 35, saved 2.2 kg of CO_2, and never opened a second app.